

CHEMISTRY CONCENTRATION

Guidelines for Using Research-for-Credit Courses to Fulfill the Advanced Laboratory Requirement

The chemistry concentration requirement for an Advanced Laboratory course (item 1.e. in the Harvard College Fields of Concentration) is designed to formalize undergraduate students' problem-solving, critical thinking, experimental design, and scientific communication abilities by encouraging active learning both inside and outside the laboratory. It can be fulfilled by taking one of the following courses: CHEM 100R, CHEM 135, CHEM 145, or CHEM 165.

Starting in the Fall 2025 term, students can petition to fulfill the Advanced Laboratory requirement with two courses of CHEM 91R/98R/99R¹ (i.e., Research-for-Credit courses) taken for **two consecutive semesters** doing **chemical research**. Students who select this option are required to submit a **scientific technical report** at the conclusion of each semester, which must receive a satisfactory evaluation from the Principal Investigator (PI).

Eligibility:

Any sophomore, junior, or senior concentrators whose previous record is satisfactory may petition to use Research-for-Credit courses to fulfill the Advanced Laboratory requirement according to these guidelines.

A petition to use Research-for-Credit courses to fulfill Advanced Laboratory requires the signature of a qualified research adviser (Harvard-affiliated PI) who must be an officer of the University, and whose professional competence is appropriate for a subfield in Chemistry. The two semesters of the Research-for-Credit courses to fulfill Advanced Laboratory requirement should be with **the same PI**.

Petition process and course timeline:

Students should discuss their petition with their PI as soon as possible, keeping in mind that a final, signed version of this petition, including the project outline, must then be submitted to the Office of Undergraduate Studies in Chemistry before the add/drop deadline of the term. Detailed submission instructions are on page 3.

The petition must include an outline of the student's research in the PI's lab. Optionally, if the PI would like to modify the instructions/specifications of the written report, these modifications should be included in the petition as well.

By the last day of the Reading Period, the student must submit to the adviser the written report that meets the report guidelines outlined in this petition. The granting of credit ("SAT" or "UNSAT") will be determined by the adviser; the Office of Undergraduate Studies in Chemistry will ask advisers to submit the grade by the fifth day of the Examination Period.

Notes for chemistry concentrators:

The research-for-credit courses under this petition are subject to the same rules for dropping and withdrawing as any other course, as described in the *Harvard College Student Handbook*. If a student is not able to complete the written report but is able to meet the expectations¹ for Research-for-Credit courses (CHEM 91R/98R/99R) according to the adviser's evaluation, then students will still receive "SAT" grade and the course will count as a Research-For-Credit course.

For a Chemistry concentrator whose petition is approved, the fulfillment of Advanced Laboratory will also count as fulfilling one of the four courses in Chemistry numbered 100 or higher, which are required for the honors track. Chemistry concentrators should still meet the total number of courses required by the concentration. In addition, students should be mindful of the credit requirements described in the *Harvard College Student Handbook*.

For Chemistry concentrators who are also admitted to the Concurrent Master's Program in Chemistry, Research-for-Credit (Chem91R/98R/99R) courses will appear on their Master's transcript as opposed to their College transcript. Students who meet the requirements above are still considered to have fulfilled the Advanced Laboratory requirement.

¹ The expectations for Research-for-Credit courses (CHEM 91R/98R/99R) if **not** used to fulfill the Advanced Laboratory requirement are at tinyurl.com/chem-research-for-credit and are summarized below:

- To be eligible to enroll, you must have a Harvard-affiliated principal investigator agree to mentor you for the semester.
- The workload averages about **10 hours** of research-related work each week for the term. Students are expected to complete 130 hours in total by the end of semester.
- Students must complete basic laboratory safety training and other safety training required by the host lab prior to starting work. Students must always follow the safety guidelines and training provided by the lab/department.
- No written work is required.
- Students will earn either a "SAT" or "UNSAT" grade determined by the PI.

Written Report Guidelines

This report should reflect the scientific and technical aspects of the research project and serve as an opportunity to develop skills in scientific communication and writing. Below are the guidelines and requirements for the report. These parameters **can be modified at the Principal Investigator's (PI) discretion**.

Report Structure and Content:

1. Title Page: Include Project Title, Student's Name, PI's Name, Course Title and Code, Date of Submission
2. Abstract: Provide a concise summary of the research, including the main objectives, methodology, key results, and conclusions. (Length: Approximately 250-300 words.)
3. Contributions: Provide a brief statement describing contributions from the student and other collaborators, if any, that are included in the report. (Length: Approximately 250 words.)
4. Introduction: Introduce the research topic and provide background information necessary for understanding the project; Clearly state the research question or hypothesis; Discuss the significance and objectives of the study. (Length: 1-2 pages.)
5. Materials and Methods: Describe the experimental design, materials, and procedures in detail to allow reproducibility; Include any analytical techniques and instrumentation used; Mention any specific safety and ethical considerations. (Length: 2-3 pages.)
6. Results: Present the findings of the research, using text, tables, and figures as appropriate; Each figure and table should be accompanied by a legend and be referenced in the text; Organize the results in a logical and coherent manner. (Length: 2-3 pages.)
7. Discussion: Interpret the results and discuss their implications in the context of existing literature; Address any limitations of the study and propose future research directions; Compare findings to previous work and highlight novel contributions to the field. (Length: 2-3 pages.)
8. Conclusion: Summarize the main findings and their significance; Restate the research question and reflect on whether it was addressed effectively. (Length: Approximately 1 page.)
9. References: List all references cited in the report in an appropriate scientific format (e.g., APA, ACS); Ensure accuracy and completeness of all references.
10. Appendices (if applicable): Include supplementary material such as raw data, calculations, or additional figures.

Formatting: 10-15 pages (excluding title page, references, and appendices); Times New Roman, size 12; 1.5 or double-spaced; Margins of 1-inch on all sides; Page numbers at the bottom of each page.

Additional Considerations:

1. Time spent writing the report does not count toward the 10 hours per week of research time required by CHEM91/98/99r; students should budget their time accordingly.
2. Ensure clarity and coherence throughout the report; Proofread the document for grammar and spelling errors; Use scientific terminology appropriately.

Resources:

1. These guidelines aim to provide a structured framework for your report while allowing flexibility for different types of projects. It is crucial to maintain open communication with your PI and your graduate student/postdoctoral mentor to ensure that the report meets the expectations and requirements of your research work.
2. <https://www.sciencewrites.org/non-empirical-papers>
3. https://writingproject.fas.harvard.edu/sites/g/files/omnuum4531/files/hwp/files/life_sciences.pdf

Petition for Using Research-for-Credit Courses to Fulfill Advanced Laboratory Requirement

INSTRUCTIONS FOR THE STUDENT: After reading the “*Guidelines for Using Research-for-Credit Courses to Fulfill the Advanced Laboratory Requirement*,” please complete the following petition in close consultation with your Chemistry Concentration Advisors (Dr. Gregg Tucci and Dr. Lu Wang at chemistrydus@fas.harvard.edu). Submit the petition to your research adviser (the principal investigator of the research group) for their approval before the add/drop deadline of the first of the two consecutive terms in which you plan to fulfill the Advanced Laboratory requirement. A final, signed version of this petition, including the project outline, must then be submitted to the Office of Undergraduate Studies in Chemistry via this Google Form (link: <https://tinyurl.com/ChemAdvLabForm>), before the add/drop deadline. You will receive the petition result in an email within two weeks of submission.

Your research adviser may refer questions about these guidelines or your petition to Gregg and Lu (chemistrydus@fas.harvard.edu).

STUDENT INFORMATION

Name:		Class:	
College Email:			
<i>I petition to count the following two consecutive terms of Research-for-Credit Courses to fulfill Advanced Laboratory requirement:</i>	☐ Fall	☐ Spring	20__
	☐ Fall	☐ Spring	20__

Please attach the following to this petition:

- Research Project:** Provide a concise summary of the research, including the main objectives and methodology. (Length: Approximately 250-300 words.) If your PI is not on the faculty of the Department of Chemistry and Chemical Biology, make sure to highlight the chemical relevance of your research project.
- Modification to the Written Report Guidelines** (if there is any): discuss the Written Report Guidelines with your PI and see whether there is any modification needed from your PI's perspective based on the nature and scope of your project.

Student's signature:		Date:	
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INSTRUCTIONS FOR THE RESEARCH ADVISER: Advisers are encouraged to consult with Chemistry Concentration Advisors (Dr. Gregg Tucci and Dr. Lu Wang at chemistrydus@fas.harvard.edu) about the Research-for-Credit Courses Fulfilling Advanced Laboratory Guidelines in advance of committing to signing this document.

RESEARCH ADVISER'S APPROVAL

I approve of this student's petition, including the research project and any modifications to the Written Report Guidelines (if applicable). I agree to evaluate the written report submission specified by the Written Report Guidelines (and any modifications, if applicable). I will submit a final grade of either "SAT" or "UNSAT," no later than one week after the first day of the final examination period, along with a brief statement about the student's work, to the link provided by the Office of Undergraduate Studies in Chemistry.

Adviser's signature:		Date:	
Adviser's name in print:		E-mail:	
Harvard address:			
Harvard telephone:			
Harvard title:			